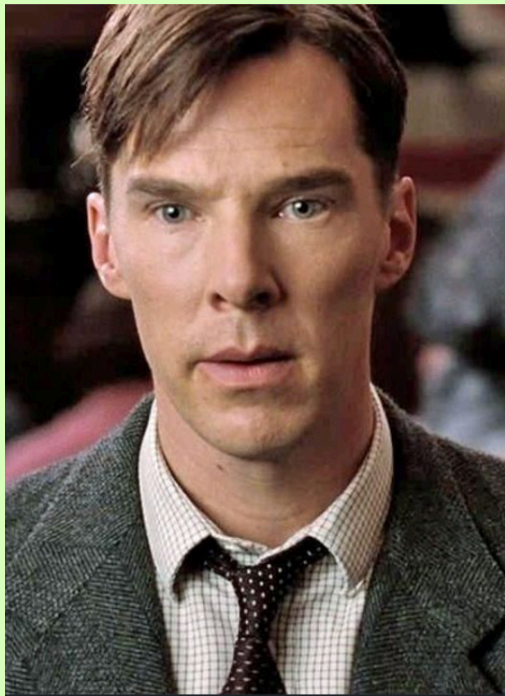




Welcome! 🍌

EECS 376, Sec. 18
D5: Computability



Sam Hiken

In class this week:

Two formal models of computing

- **DFAs:** computing without memory
- **Turing machines:** computing with memory

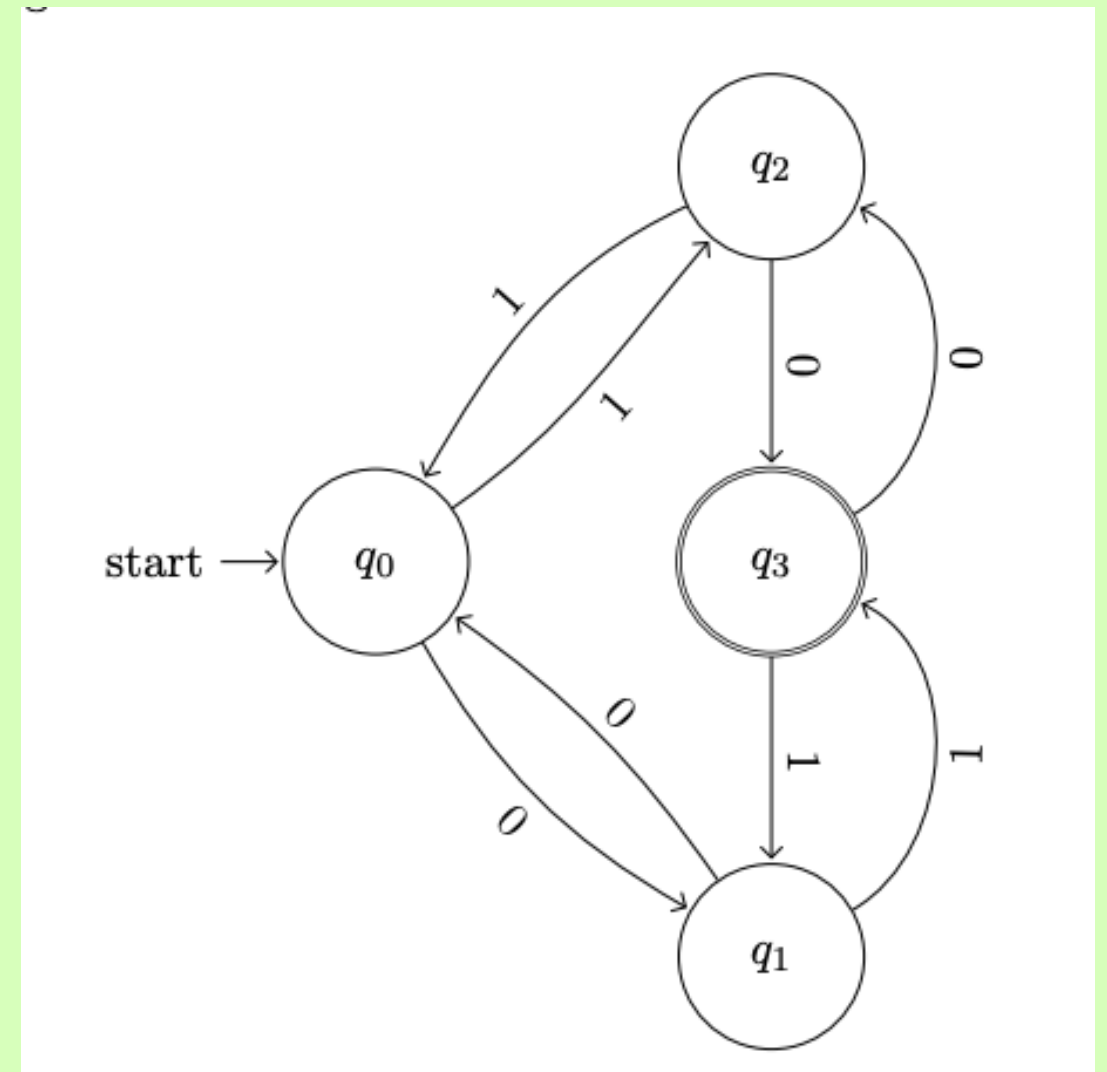
DFA

Deterministic Finite Automata

Consist of:

- alphabet
- states (including starting and accept states)
- transition function (which state to move to)

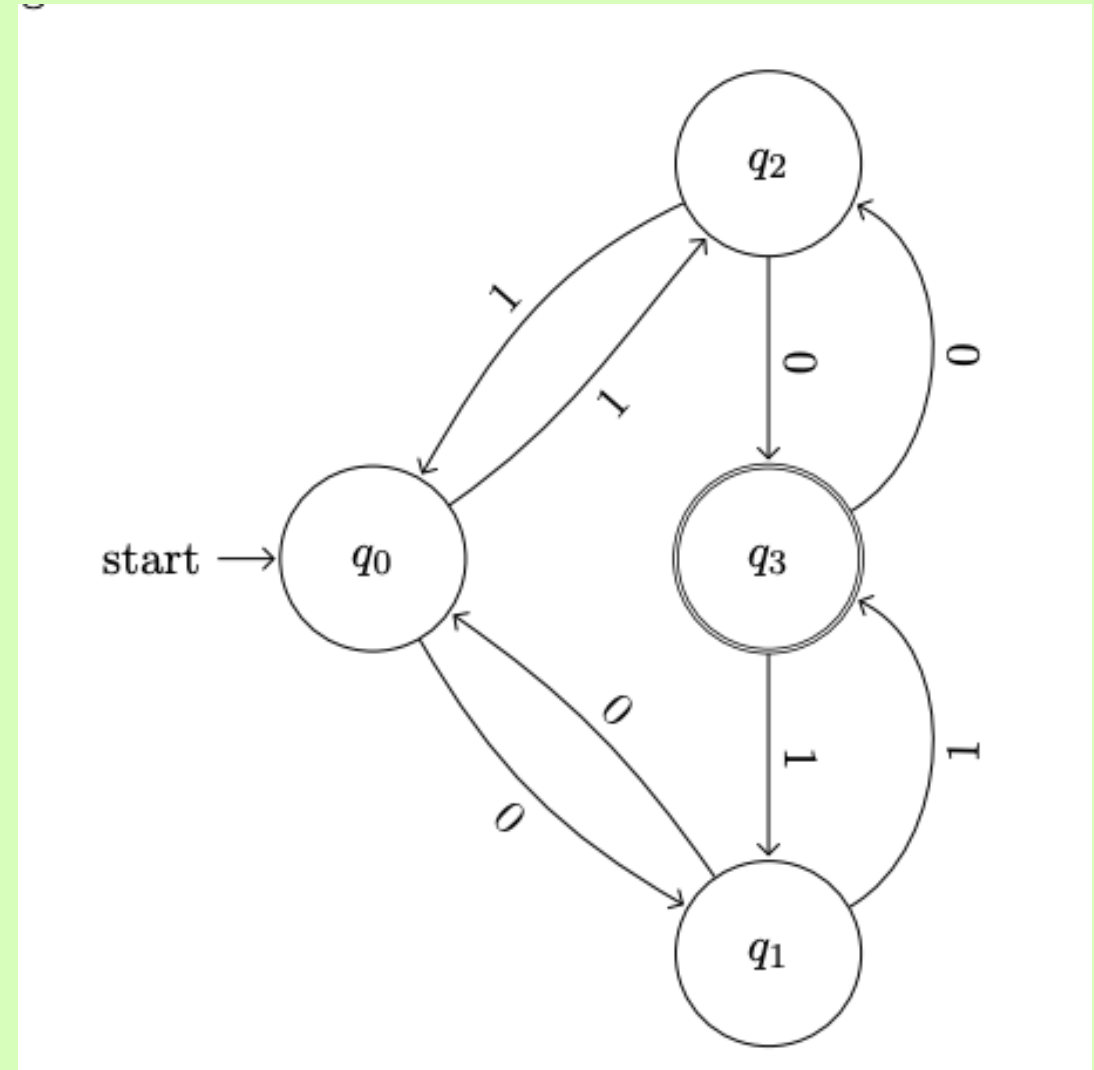
A string is *accepted* by a DFA if it traces a path from the start state to an accept state.



DFA

Example

1110 is accepted by this DFA



$$q_0 \rightarrow q_2 \rightarrow q_0 \rightarrow q_2 \rightarrow q_3$$

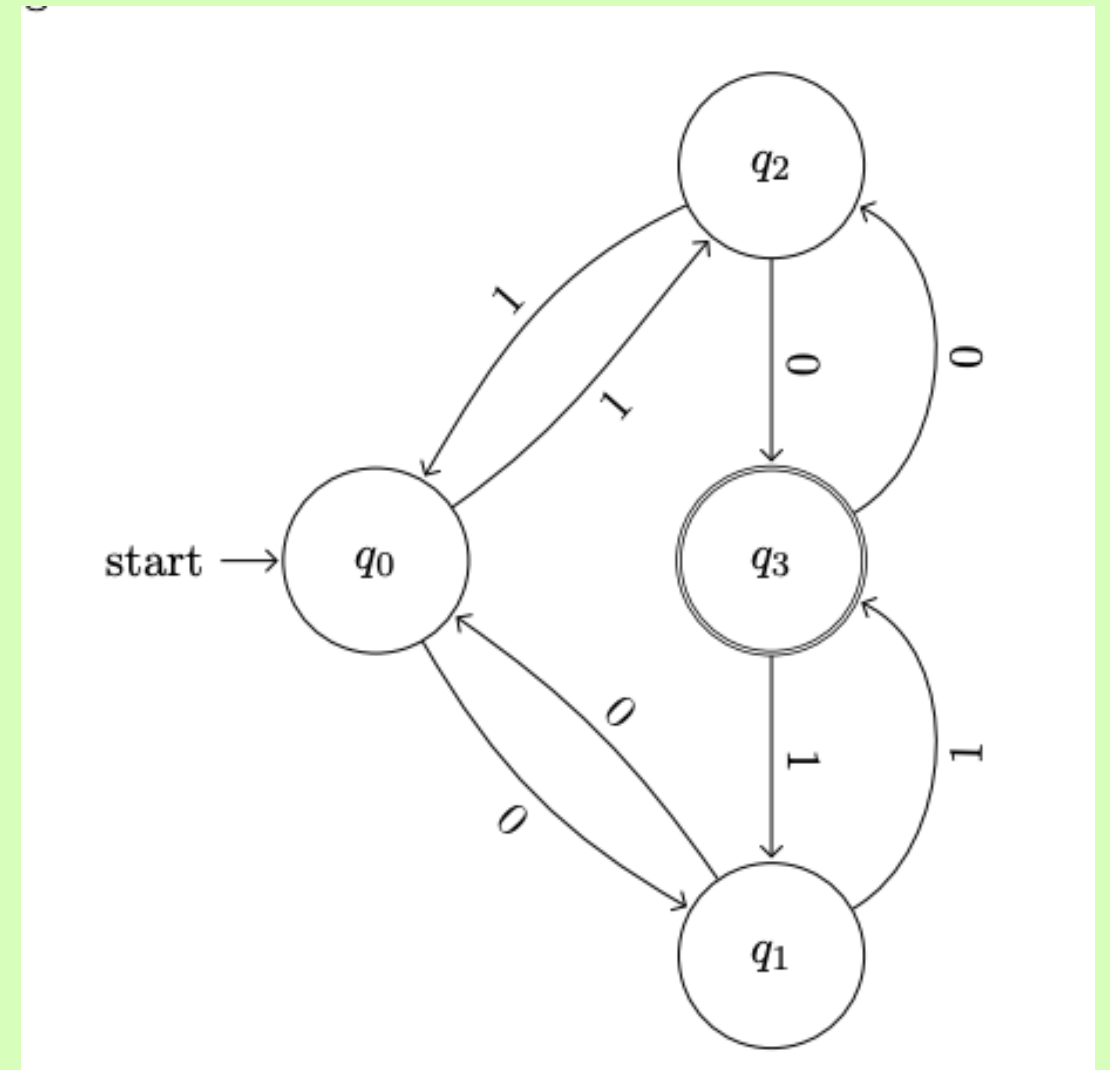
DFA's

Quiz:

Which of the following strings is NOT accepted by this DFA?

- 0001
- 11000

• 0011

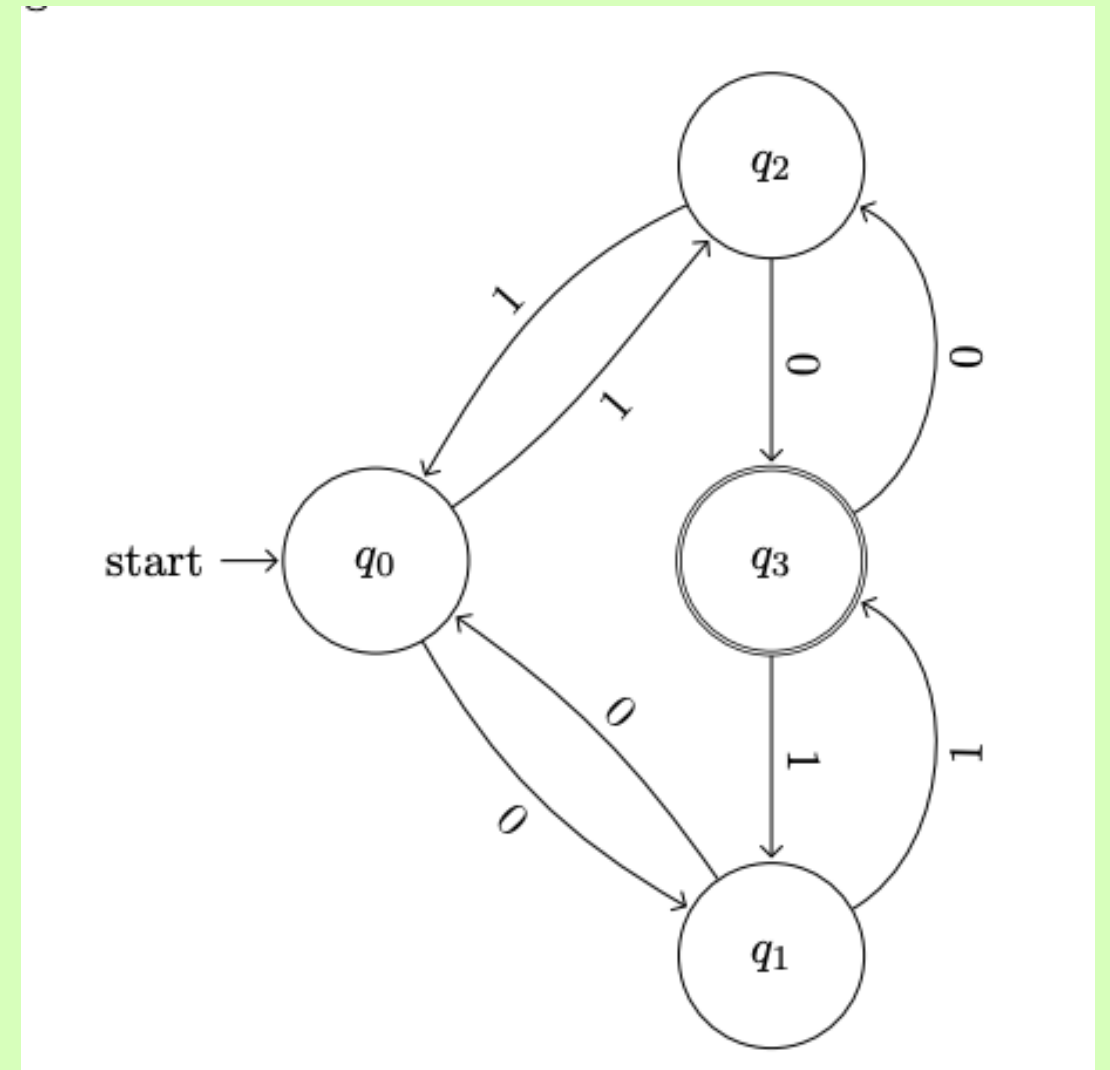


DFAs

Note: DFAs represent *computing without memory*

The machine can only see the current character it's looking at

How do we model computing with memory?



Turing Machines

Like a DFA, but you also have unbounded memory!

- memory = infinite tapes

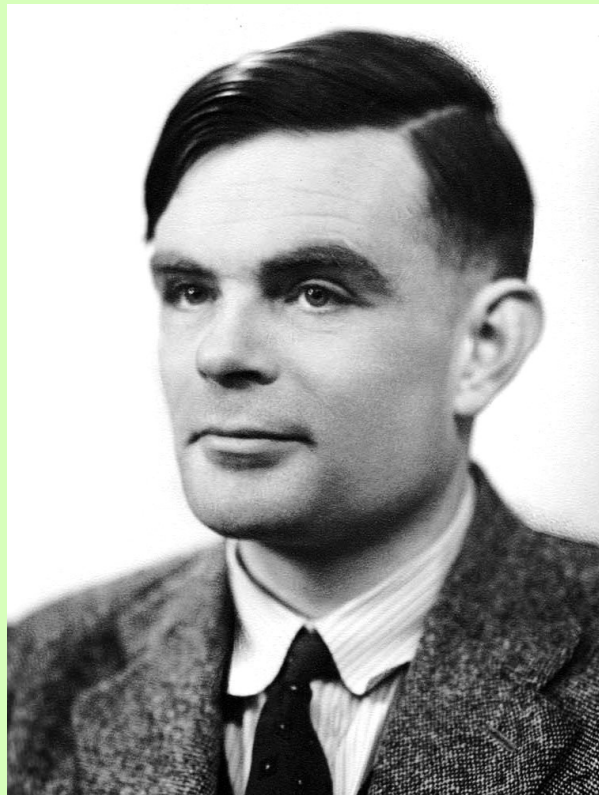
Like before, you have

- alphabet
- states (including starting state + accept states)
- transition function

But now, the transition function reads and writes to the tapes!

Turing Machines

Where did this idea come from?



← Alan Turing

Alan Turing wanted to define the intuitive notion of an “algorithm” in a mathematically rigorous way.

Turing Machines

Church-Turing thesis:

a problem can be solved by an “algorithm” if and only if it can be solved by a Turing machine

Hence, in this class, in order to prove that a problem can be solved by a Turing Machine, it will suffice to provide pseudocode

Gather into groups of 2-3, work on the worksheet

Worksheet link:

<https://tinyurl.com/yynx4mfy>

[this is in the class Google Drive]

